# FastSAM-CD: Remote Sensing Image Change Detection Using Vision Foundation Models With Stronger Encoder and Decoder

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***Abstract*—Remote sensing image change detection (RSICD) is a crucial technique for Earth observation. However, the mainstream RSICD methods still face two main challenges. First, the encoding stage often fails to capture fine-grained structural representations, particularly in scenarios involving cross-scale targets and complex boundaries. Second, the decoding stage lacks effective modeling of spectral heterogeneity and direction-sensitive channel interactions, which severely limits the ability to accurately recognize strip-shaped objects. To address these issues, this article proposes a network for RSICD named fast segment anything model (FastSAM) -change detection (CD), which extends FastSAM with two dedicated modules. First, we design a hyperfusion multiview adapter (HFM Adapter) for the encoder of FastSAM-CD. It significantly enhances the model's ability to capture fine-grained boundaries across scales by constructing multiview paths in both spatial–channel and local–global dimensions. Second, we propose a spectral–axial dynamic modulation module (SDM Module) for the decoder. It enhances the decoder's extraction of spatial–frequency-domain information through axial sensing and frequency-domain analysis, significantly improving the model's detection accuracy for strip-shaped objects. The experiments on three RSICD datasets demonstrate that the proposed method outperforms the existing mainstream RSICD methods in terms of accuracy and efficiency.**

***Index Terms*—Change detection, fast segment anything model (FastSAM), hyperfusion multiview adapter (HFM Adapter), spectral–axial dynamic modulation module (SDM Module).**

Received 30 September 2025; revised 7 January 2026; accepted 28 January 2026. Date of publication 3 February 2026; date of current version 20 February 2026. This work was supported in part by the National Natural Science Foundation of China under Program 62271296 and Program 62201334; in part by the Innovation Capability Support Plan Project in Shaanxi Province under Grant 2025RS-CXTD-012; in part by the Scientific Research Program Funded by Shaanxi Provincial Education Department under Grant 25JP023, Grant 23JP014, and Grant 23JP022; and in part by the Nature Science Basic Research Program of Shaanxi under Grant 2025JC-YBQN-800. *(Corresponding author: Tao Lei.)*

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Digital Object Identifier 10.1109/TGRS.2026.3660946

## I. Introduction

REMOTE sensing image change detection (RSICD) is a key technology for monitoring dynamic changes on the Earth's surface [1]. The technology has been widely applied to many fields such as environmental monitoring, urban planning, and disaster response [2]. Traditional RSICD methods rely mainly on pixel-level or object-level comparison techniques such as differential analysis [3] and principal component analysis [4]. However, these traditional methods depend heavily on manual feature extraction, which makes it difficult to effectively represent the semantic information of remote sensing images (RSIs), resulting in low change detection (CD) accuracy for complex remote sensing scenarios [5], [6]. With the development of deep learning technology, neural network-based methods have gradually become mainstream for RSICD tasks. Convolutional neural networks (CNNs) have made significant progress in a variety of visual tasks, thanks to their powerful feature extraction capabilities [7], [8]. In the RSICD field, CNN models can automatically learn multiscale feature representations of input images in an end-to-end manner, thereby effectively improving the accuracy of CD tasks [9]. However, traditional CNN-based architectures are limited by fixed receptive fields, making it difficult to model long-range context dependencies, which makes it hard to fully capture the spatial correlation features of cross-regional changes in high-resolution RSIs, resulting in lower detection accuracy for regions with large change ranges [10], [11]. In recent years, the Transformer-based architecture has gained extensive attention for its outstanding ability to model long sequences and has been introduced into a variety of computer vision tasks [12]. Although the Transformer is good at capturing long-range dependencies, it still faces difficulties in cross-scale structural alignment in RSIs and low efficiency in processing large-scale data, which limits its application in resource-constrained RSICD scenarios [13].

In recent years, the emergence of vision foundation models (VFMs) has brought new opportunities for RSI processing tasks. Models such as the segment anything model (SAM) [14] and its variants fast segment anything model (FastSAM) [15] are pretrained on large-scale data. They can be fine-tuned

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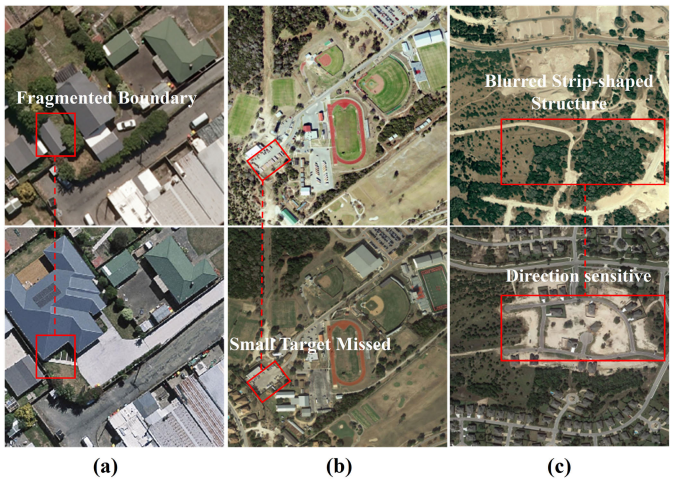



Fig. 1. Key challenges in fine-grained RSICD when applying foundational models like FastSAM. (a) Complex geometric changes suffer from boundary fragmentation due to structural misalignment (fragmented boundary). (b) Small-scale changes are missed due to cross-scale semantic misalignment (small-target missed). (c) Strip-shaped structures exhibit blurred boundaries due to a lack of spectral–axial modeling (blurred strip-shaped structure).

with a small amount of labeled data and thus adapt quickly to downstream tasks. However, SAM is mainly trained for natural image scenes, and the segmentation results depend on the accuracy of the prompts. In the RSICD task, targets tend to have weak boundaries and small interclass differences, making it difficult for SAM to generate precise change regions.

Applying FastSAM directly to high-resolution RSICD tasks also causes many challenges [16], [17]. Specifically, since FastSAM relies on CNNs for local feature extraction, its fixed receptive field limits its adaptability to regions with scale changes, especially in scenarios with significant scale changes or structural misalignment, making it difficult to achieve co-modeling of global and local structures, thereby weakening FastSAM's ability to represent fine structures such as small-scale change targets and complex boundaries. As shown in Fig. 1(a), complex geometric changes suffer from boundary fragmentation due to structural misalignment. As shown in Fig. 1(b), small-scale changes are often missed due to cross-scale semantic misalignment. At the same time, FastSAM lacks an explicit modeling mechanism for the correlation semantics among channel features during the feature reconstruction stage, resulting in insufficient channel information interaction in complex RSI contexts. In addition, although the decoder retains some detailed information through skip connections, when dealing with strip-shaped objects, due to the heterogeneity of the image spectral distribution and the large number of direction-sensitive structures, the model causes problems such as blurring of long-range semantic separation and weakening of boundaries in RSIs [18]. As shown in Fig. 1(c), strip-shaped structures exhibit blurred boundaries.

To address these issues, this article proposes a network architecture for RSICD tasks, named FastSAM-CD. It enhances the collaborative perception of structural information in RSIs through the hyperfusion multiview adapter (HFM Adapter). Furthermore, we design the spectral–axial dynamic modulation module (SDM Module) to dynamically modulate the potential correlation structure between channels in RSIs. The accuracy and efficiency of CD improve with only a small increase in network parameters. The main contributions of this article can be summarized as follows.

1) *Overall Framework*: A network architecture, FastSAM-CD, is proposed for RSICD tasks. Compared with the FastSAM architecture, this proposed architecture enhances the multiview domain structure perception ability of the encoder and the joint spectral-axial modeling ability of the decoder. It addresses the insufficiency of traditional foundation model-assisted RSICD networks in representing fine structures such as cross-scale small targets, complex boundaries, and strip-shaped features. It not only improves the detection accuracy in RSICD tasks but also enhances the overall computational efficiency of the network.
2) *Encoding Stage*: An HFM Adapter is designed to enhance the FastSAM-CD encoder's ability to model multiscale information in RSIs. Based on fine-tuning a small number of learnable parameters, the Adapter employs multiscale grouped convolution and attention mechanisms to achieve multiview collaborative perception of features from both local–global and spatial–channel dimensions, addressing the encoder's insufficient ability to represent fine-grained boundaries caused by cross-scale semantic misalignment and geometric misalignment in RSIs.
3) *Decoding Stage*: An SDM Module is designed to integrate the frequency-domain distribution pattern and direction-sensitive structural information of channel features in the decoding stage. This module achieves dynamic modulation of the spatial domain and frequency domain structure through the joint modeling of spectral decomposition and axial sensing, solving the problem of the decoder's weakening of strip-shaped feature representation due to the heterogeneity of cross-channel spectral distribution and the large number of direction-sensitive structures in RSIs. It is particularly suitable for target types with significant directional and spectral differences, such as roads and farmlands.

## II. Related Works

### A. RSICD With Deep Learning

In recent years, deep learning techniques have made significant progress in RSICD tasks, particularly in dealing with challenges such as complex backgrounds, multiscale structures, and fine-grained targets. The current mainstream methods can be roughly classified into three categories: CNN-based methods, Transformer-based methods, and hybrid architectures of CNN–Transformer, emphasizing local feature extraction, global dependency modeling, and the complementary fusion of features between the two. Recently, Mamba [19] has emerged as a promising alternative to Transformers, demonstrating strong efficiency in long-range sequence modeling for RSICD tasks [20], [21], [22]. This architecture

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may provide a new paradigm for balancing global dependency modeling and computational efficiency.

CNN-based methods, which focus on local feature extraction, dominated early RSICD research. Spatialtemporal attention neural network (STANet) [23] introduces spatiotemporal attention mechanisms, incorporating a multiscale self-attention module in feature extraction to capture cross-spatiotemporal dependencies between pixels of an image. Dual-branch multilevel intertemporal network (DMINet) [24] proposes a joint attention module to unify self-attention and cross-attention and enhances the response to changed regions through multilevel difference aggregation. Based on this, edge-guided recurrent convolutional neural network (EGRCNN) [25] improves the recognition accuracy of building boundary changes by fusing structural prior information with depth features through an edge-guided mechanism. Furthermore, deeply supervised attention metric-based network (DSAMNet) [26] combines deep supervision with attention mechanisms and introduces metric modules to enhance feature discrimination. Recently, adaptive deformable multi-scale network (ADMNet) [27] proposes a Siamese network for building CD. It employs adaptive deformable convolution modules to capture irregular building edges. It also uses a multilevel progressive feature fusion module driven by large-model transfer learning to aggregate multiscale contextual and edge cues, thereby improving building-level change localization. In addition, both dual task siamese convolutional network (DT-SCN) [28] and fully convolutional early fusion (FC-EF) [29] further introduce the dual-task constraint mechanism and improve the loss function, enhancing the integrity and robustness of CD at the object level.

Transformer-based methods, which excel in modeling long-range dependencies, demonstrate superior performance in CD tasks. Bitemporal image transformer (BIT) [30] compresses the image representation into a small number of semantic labels and uses a Transformer to model spatiotemporal context efficiently, significantly improving detection efficiency and accuracy. ChangeFormer [31] fuses the hierarchical Transformer encoder with the multilayer perceptron (MLP) decoder into the Siamese network, which can precisely capture multiscale change information. Multi-scale feature aggregation via transformer network (MFATNet) [32] introduces spatial semantic markers as well as intraclass and interclass attention modules to achieve fine-grained semantic modeling and feature aggregation. Transformer-based context information aggregation network (TCIANet) [33] emphasizes context aggregation and edge structure modeling, and improves the extraction accuracy of target edge information through the contour graph inference module. In addition, RFormer [34] proposes direction-sensitive position coding and rotation-invariant modeling for the uncertainty of imaging angles in RSI, effectively enhancing the robustness of the model.

Hybrid architectures of CNN–Transformer, which aim to balance local detail capture and global semantic modeling, have become a prevalent design in RSICD. CTFINet [35] proposes a linear cross-attention mechanism to enhance cross-temporal feature interaction and combines affine alignment to optimize spatial consistency of multiscale features. HSSENet [36] develops an efficient backbone network that combines the long-range modeling capabilities of Transformers and the local extraction capabilities of CNNs, refining the representation of differential features through a two-level fusion strategy. EATDer [37] uses Siamese encoders, with each branch containing three adaptive vision converters, which can effectively extract spatiotemporal features in bi-temporal images and further enhance the localization accuracy of the changed regions through an edge-aware decoder.

Overall, the deep learning-based RSICD methods perform well in modeling the overall semantic features of changed regions, but they still have limitations in dealing with small targets and edge details.

### *B. RSICD With the Assistance of VFMs*

In recent years, VFMs have made significant progress with the accumulation of large datasets and the development of self-supervised learning techniques [14]. As a powerful model, SAM demonstrates zero-shot generalization on unseen objects and scenes by combining Vision Transformer encoders [38] with prompt mechanisms. SAM's prompt mechanisms include sparse prompts (dots, boxes, and text) and dense prompts (masks) [39], enabling it to generate semantic-level segmentation masks based on different input patterns [40], [41]. FastSAM [15] further enhances inference speed while ensuring segmentation accuracy through lightweight network design, expanding the deployment capability of VFMs in resource-constrained scenarios.

In the RSICD domain, RSPrompter [42] is an instance segmentation model based on SAM that better adapts to complex scenes in RSI by automatically generating semantically relevant prompt embeddings. Similarly, the RSAM-Seg [43] model uses adapters to learn domain-specific knowledge, introduces two scaling modules in the multihead attention module of the SAM encoder, and inserts Adapter-Feature to enhance the ability of modeling high-frequency image details. In addition, MeSAM [44] is a multiscale enhanced SAM model. It introduces an initial mixer in the image encoder to retain high-frequency features and designs a mask decoder with remote sensing correction capabilities to improve the model performance for RSI segmentation tasks. SAM-CD [45] uses a frozen FastSAM encoder and proposes a convolutional adapter to aggregate change information, in addition to introducing semantic learning branches to model the underlying semantic information in RSI. ASS-CD [46] adapts FastSAM and Swin-Transformer [47] to the RSICD task by introducing a convolutional attention module, which reduces interference from complex backgrounds and enhances the focus on multiscale objects. At the same time, a hierarchical deep supervision module is used to apply deep supervision to multiscale feature maps [48], [49]. Semantic change detection (SCD)-SAM [50] applies SAM to semantic CD by integrating a MobileSAM branch and a CNN branch into a contextual semantic change-aware dual encoder. This design incorporates depth feature interaction to effectively couple global and local

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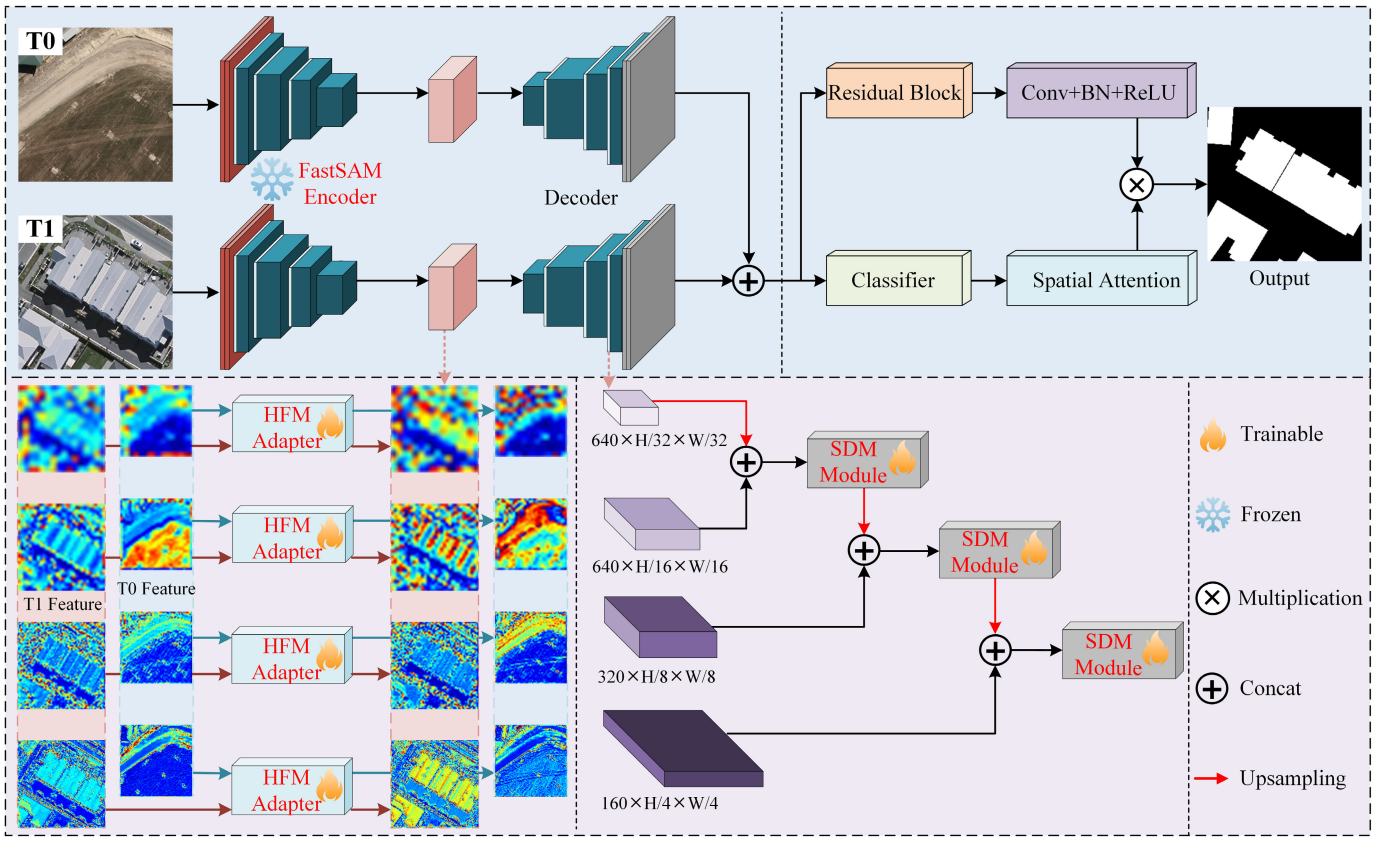



Fig. 2. Network of the proposed FastSAM-CD. First, the FastSAM encoder extracts feature maps from the input bi-temporal images. An HFM adapter is inserted after the FastSAM encoder to enhance cross-scale semantic modeling and long-range structural perception. Subsequently, an SDM module is embedded in the decoder to enhance the representation of spatial-domain structures and frequency-domain patterns in RSIs.

representations. The above VFM-based methods have been introduced into the RSICD task, but they are still inadequate in perceiving fine-grained varying structures and modeling complex scenarios in RSIs [51].

Overall, although the CNN-based methods perform well in local feature extraction, they are limited by the fixed receptive field and perform poorly in depicting small-scale targets and complex boundary structures. Transformer-based methods have the advantage of long-range modeling, but they face high computational overhead and difficulties in aligning RSI across scales. Although VFM-assisted methods have zero-shot generalization capabilities, they cannot perceive fine-grained changes in RSIs and model spectral structure features.

To overcome these challenges, this article proposes an RSICD network framework based on FastSAM, named FastSAM-CD. It collaboratively enhances the model's cross-scale structural perception ability and spectral semantic expression ability.

## III. Proposed Method

### A. Overall Framework

Our proposed FastSAM-CD framework for RSICD is shown in Fig. 2. The framework mainly consists of three parts: a frozen FastSAM encoder, a multistage decoder, and a final pixel-level CD module.

In the encoding stage, the model uses the frozen FastSAM encoder to extract multiscale features of bi-temporal images, respectively. After the FastSAM encoder, a trainable HFM Adapter is introduced to enhance feature representation. By constructing 2-D multiview-domain paths, the HFM Adapter can transform the base features output by the FastSAM encoder into a high-order perception feature, thereby significantly enhancing the model's ability to detect fine-grained boundaries and small objects in RSIs. The related design will be explained in Section III-B. In the decoding stage, the model uses a multistage decoder to gradually restore the encoded features while introducing the proposed SDM Module, which enhances the representation of spatial-domain structures and frequency-domain patterns in RSIs. The related design will be explained in Section III-C.

The backbone network of the FastSAM encoder is YOLOv8-seg [52], with efficient feature extraction through the CNN and multiscale feature fusion using spatial pyramid pooling-fast. Then the two semantic feature maps $s_1$ and $s_2$ can be expressed as

$$s_1, s_2 = \mathcal{D}\left(\mathcal{E}\left(I_A, I_B\right)\right) \tag{1}$$

where $(I_A, I_B)$ represents the bi-temporal images, $\mathcal{E}(\cdot)$ represents the encoder of our proposed FastSAM-CD, and $\mathcal{D}(\cdot)$ represents the decoder of FastSAM-CD.

To facilitate the network's learning on changed regions, we perform feature fusion on semantic features extracted from

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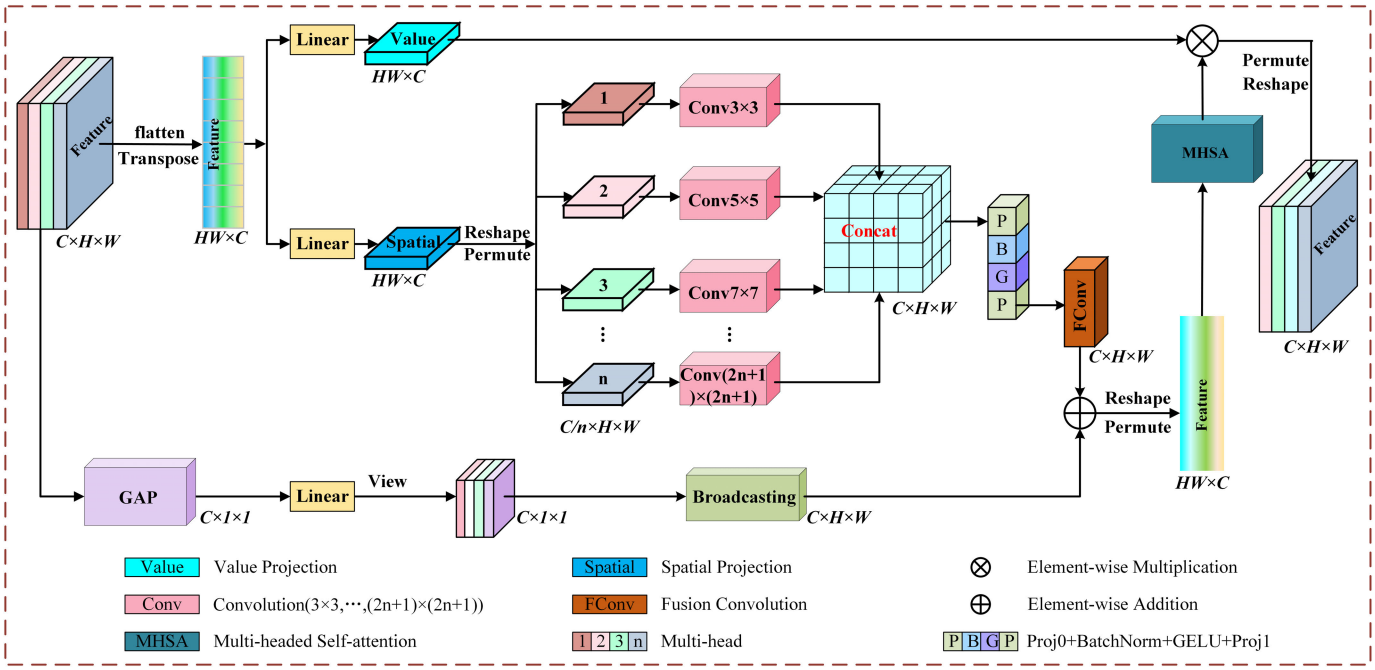



Fig. 3. Illustration of our HFM adapter. This module is inserted after the FastSAM encoder and is primarily composed of four-branch multiscale grouped convolutions, channel reconstruction, channel awareness, and a multihead self-attention mechanism.

decoder outputs. Specifically, the two semantic feature maps $s_1$ and $s_2$ are concatenated and passed through a residual block to capture cross-temporal interactions. The fused representation $F$ is then refined by a $1 \times 1$ convolution, batch normalization (BN), and rectified linear unit (ReLU) activation to produce enhanced semantic features $F'$

$$F = \text{ResBlock}\left(\text{Concat}\left(s_1, s_2\right)\right) \tag{2}$$

$$F' = \text{ReLU}\left(\text{BN}\left(\text{Conv}_{1\times1}\left(F\right)\right)\right). \tag{3}$$

Meanwhile, $s_1$ and $s_2$ are concatenated and fed into a classifier to generate an initial change map $D$, which is then used to compute spatial attention weights $W_d$

$$D = \text{Classifier}\left(\text{Concat}\left(s_1, s_2\right)\right) \tag{4}$$

$$W_d = \sigma\left(\text{SA}\left(D\right)\right) \tag{5}$$

where SA denotes the spatial attention module, $\sigma$ is the sigmoid activation function, and $W_d \in [0,1]^{H\times W\times 1}$ serves as spatial attention weights to highlight changed regions.

Subsequently, the enhanced semantic features $F'$ are dynamically weighted by $W_d$ to generate the output feature map $F_{\text{att}}$

$$F_{\text{att}} = F' \odot W_d \tag{6}$$

where $\odot$ denotes element-wise multiplication. Finally, $F_{\text{att}}$ is passed through a prediction head to generate the final CD results.

### B. HyperFusion Multiview Adapter

FastSAM shows excellent generalization ability for natural images, but its encoder suffers from structural limitations when applied to RSICD tasks. The RSI is characterized by significant scale differences, high resolution, and diverse forms of changed regions [53], which leads to significant differences in feature modeling requirements compared to natural images. Existing direct transfer strategies based on FastSAM generally suffer from the following problems [54]: FastSAM encoder is limited by local convolutional receptive fields and has difficulty effectively capturing cross-scale and cross-regional context relationships among remote sensing targets. The serial encoding path limits the semantic fusion ability, which can easily lead to incomplete representation of changed regions in RSIs. In addition, the pretrained model is dominated by the general semantics of natural images, resulting in the encoder's insufficient response to weak boundaries and small-sized targets in RSIs.

To address these issues, this article proposes an HFM Adapter, which is inserted after the FastSAM encoder, as shown in Fig. 3. It aims to introduce a 2-D modeling mechanism which is across scales and channels without damaging the backbone structure. The HFM Adapter models from two dimensions. On the one hand, it builds a spatial–channel interaction path to alleviate the convolution receptive field limitation and improve the encoder's ability to model fine-grained structures. On the other hand, a local–global parallel modeling mechanism is designed to enhance the encoder's ability to represent the context of changed targets through the collaborative modeling of short-range attention and long-range perception. Compared with the strategy of directly fine-tuning the encoder, the HFM Adapter has greater migration flexibility and structural adaptability, which can significantly improve the semantic alignment and change perception ability of FastSAM in remote sensing scenarios.

*1) Local Paths:* The changed targets in RSIs often have characteristics such as small scale, weak boundaries, and fragmented structure, which require the model to have a stronger ability to model local features. To this end, we

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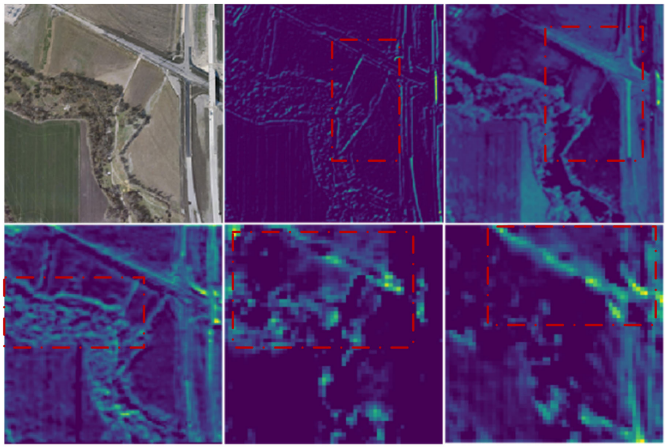


Fig. 4. (Top-left) Original image and the visualization heatmaps of the outputs from different convolutional heads in each group.

design a local path based on multiscale grouped convolution, which enables the model to adaptively capture local changed structures at receptive fields of different scales. First, the input features $X'_{\text{spatial}} \in \mathbb{R}^{C\times H\times W}$ are divided into $n$ groups in the channel dimension, with each group corresponding to a convolution head, $X_i \in \{X_1, X_2, X_3, \ldots, X_n\}$, where $X_i \in \mathbb{R}^{C/n\times H\times W}$ represents the input of the $i$th convolution head, and each convolution head uses depthwise separable convolution kernels of different receptive fields to capture information at a specific scale in parallel. To capture features at different scales, the HFM Adapter uses $n$ depthwise separable convolutions ($n = 4$ in experiments) with convolution kernel sizes of $3 \times 3$, $5 \times 5$, $7 \times 7$, and $9 \times 9$, respectively. These four receptive fields of convolution kernels can effectively cover small, medium, and large-scale targets, ensuring complete information while avoiding redundant modeling. In all these depthwise separable convolutions, a kernel size of $k_{\text{conv}} \times k_{\text{conv}}$ is used with a stride of 1 and padding of $\lfloor k_{\text{conv}}/2 \rfloor$, so that the spatial resolution of the feature maps is preserved in each branch. The output $S_i$ of each convolution head is calculated as follows:

$$S_i = \text{GELU}\left(\text{BN}\left(\text{DWConv}_i\left(X_i\right)\right)\right) \tag{7}$$

where $\text{DWConv}_i$ represents the $i$th depthwise separable convolution and BN represents the BN layer. Then, all of $S_i$ are concatenated to form the initial fused feature representation

$$S_{\text{out}} = \text{Concat}\left(S_1, S_2, \ldots, S_n\right) \tag{8}$$

where $S_{\text{out}} \in \mathbb{R}^{C\times H\times W}$ represents the result of multiscale feature fusion.

Fig. 4 is the visual heat map of the multiscale grouped convolution stage. It can be found that each convolution feature map focuses on granularity features at different levels, proving that the ability to capture details at different levels has been improved.

Although multiscale grouped convolution can enhance the representation of local spatial receptive fields, the grouping operation leads to insufficient interaction among the features of channels at different scales. To this end, we introduce channel reconstruction mechanisms in local branches. Specifically, this mechanism first increases the channel dimension to enhance the coupling interaction of features in high-dimensional space. Low-rank projection is then carried out through channel compression operations to suppress redundant information and strengthen critical channel responses. Through a nonlinear mapping process of dimension expansion and dimension reduction, the isolation between channels is broken. This operation enhances the selective representation and adaptive reconstruction of cross-scale semantic information in the channel dimension compared to the straight channel connection method in FastSAM

$$S_{\text{exp}} = \text{GELU}\left(\text{BN}\left(\text{Proj}_0\left(S_{\text{out}}\right)\right)\right) \tag{9}$$

$$S_{\text{fuse}} = \text{FConv}\left(\text{Proj}_1\left(S_{\text{exp}}\right)\right) \tag{10}$$

where $S_{\text{exp}}$ represents the expanded feature, $S_{\text{fuse}}$ represents the fused feature, $\text{Proj}_0$ and $\text{Proj}_1$ represent linear projections for channel expansion and reconstruction, respectively, and FConv represents a fusion convolution operation.

The local path efficiently extracts the local spatial features through multiscale grouped convolution and enhances the fluidity of local information in the channel dimension by using the channel reconstruction mechanism. However, FastSAM encoder still has certain limitations in global context capture, especially the lack of an effective context fusion mechanism when dealing with cross-scale features in RSIs. To make up for this, we design a global path.

*2) Global Paths:* Compared with spatial modeling, channel modeling focuses more on screening features of different semantic categories. In RSIs, due to characteristics such as small-target areas and blurred boundaries, low response amplitude and high semantic sparsity are often shown in the channel dimension, which makes it difficult for the network to assign sufficient attention weights to them. To this end, we perform global channel sensing on the input feature map to capture the global features of channels. Specifically, through the global mean pooling operation and using the fully connected layer for transformation, it is represented as

$$G = W_g \cdot \text{GAP}\left(X\right) + b \tag{11}$$

where $W_g$ is the weight of the fully connected layer, GAP(X) is the global average pooling operation applied to the feature map X, $b$ is the bias term, and $G$ is the channel-wise global average pooling output. Subsequently, $G$ is added element-wise to the local features $S_{\text{fuse}}$ by broadcast, so that the final feature $S_{\text{final}}$ contains both local multiscale information and global structural information

$$S_{\text{final}} = S_{\text{fuse}} + \text{Broadcast}\left(\text{View}\left(G; C \times 1 \times 1\right)\right). \tag{12}$$

Although the local path enhances the model's ability to capture fine-grained structures, relying on such local modeling is insufficient to address the challenges of RSIs. This is because the changed targets in RSIs are often spatially scattered and structurally diverse, while maintaining semantic homogeneity. The local convolutional structure in the FastSAM encoder makes it difficult to model such long-range dependencies across receptive fields, resulting in semantic disconnection and structural fragmentation in the changed regions during the encoding stage, further weakening the model's ability to perceive large-scale targets. To this end, we further introduce

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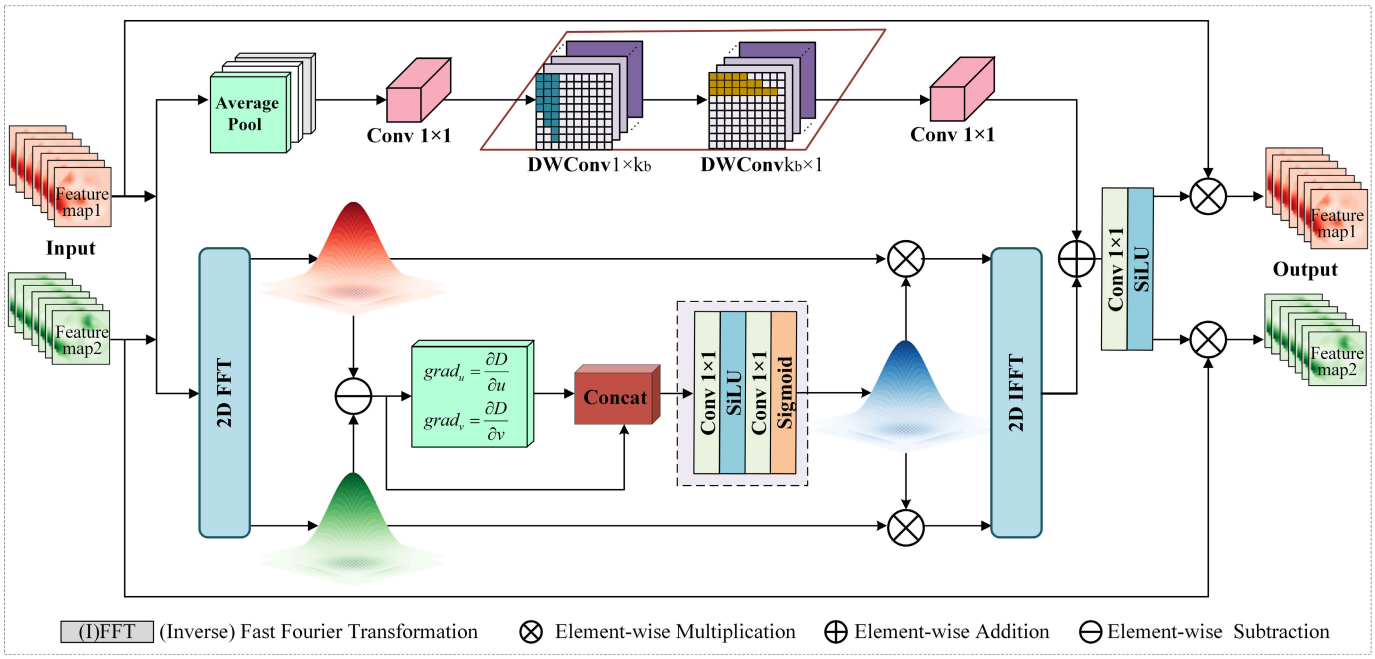



Fig. 5. Illustration of our SDM module. The upper branch models local structure via axis-wise convolution, while the lower branch captures global frequency correlations via the FFT.

the MHSA mechanism to model spatial structure dependencies by learning the explicit relationship between any two positions in input feature maps. First, $S_{\text{final}} \in \mathbb{R}^{C\times H\times W}$ is reshaped to $S'_{\text{final}} \in \mathbb{R}^{(HW)\times C}$, and then calculate the query, key, value matrices

$$Q, K, V = S'_{\text{final}} W_Q, S'_{\text{final}} W_K, S'_{\text{final}} W_V \tag{13}$$

where $W_Q$, $W_K$, and $W_V$ are learnable projection matrices in $\mathbb{R}^{C\times C}$ that project the $S_{\text{final}}$ into the query, key, and value matrices. We adopt the MHSA mechanism with $h$ heads (set to $h = 4$ in our experiments). The projected embeddings are reshaped into $h$ heads, each operating in a $d_k$ dimension subspace, where $d_k = C/h$ denotes the per-head key/query dimension. The attention output is calculated by using the generated $Q, K, V$

$$S_{\text{MHSA}} = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V. \tag{14}$$

The final spatial features $S_{\text{MHSA}}$ are multiplied by the original channel features $X_{\text{value}}$ element by element to make the $X_{\text{value}}$ more adaptable to different spatial positions. The operation can enhance the model's perception of global information, which is expressed as

$$X_{\text{out}} = S_{\text{MHSA}} \odot X_{\text{value}}. \tag{15}$$

The final output $X_{\text{out}}$ can adaptively fuse the long-range feature information provided by MHSA.

The HFM Adapter constructs two sets of functionally complementary hyperfusion multiview paths through a 2-D path design of "space–channel × local–global," effectively enhancing FastSAM's ability to represent cross-scale changes and complex-structured targets in RSIs.

### C. Spectral–Axial Dynamic Modulation Module

Conventional decoders have obvious expression flaws when dealing with strip-shaped features and direction-sensitive targets that are abundant in RSIs. First, the heterogeneity of spectral features in RSIs easily leads to channel semantic overlap and blurred boundaries, while strip-shaped features (such as roads, rivers, and building edges) rely precisely on clear, continuous boundaries and structural coherence. Therefore, such overlap severely compromises their integrity, resulting in fractures, misalignments, or incomplete recognition. Second, traditional channel attention mechanisms typically rely on global average pooling or local convolutions for channel weight reconstruction. This static or coarse-grained modeling approach struggles to dynamically adjust channel responses based on different input images, which is particularly critical for direction-sensitive targets. This is because features from different directions may activate distinct spectral response patterns, and static attention cannot flexibly capture these changes, thereby weakening the model's ability to discern directional information.

To address these issues, this article proposes an SDM Module, which enhances the decoder's ability to model the channel dimension. As shown in Fig. 5, the module incorporates a joint modeling mechanism combining frequency-domain transformation and axial sensing. This mechanism decouples and models the direction-sensitive structures and spectral heterogeneous distributions common in RSI. Furthermore, we construct a dynamic sensing mechanism by combining the spatial domain and the frequency domain.

*1) Spatial Domain:* Inspired by the idea of context-aware attention [55], we gradually increase the receptive field by adjusting the size of the convolution kernel and the stride. Specifically, we introduce a two-directional strip convolution operation. The operation uses $1 \times k_b$ depthwise separable

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convolution in the width direction and then uses $k_b \times 1$ depthwise separable convolution in the height direction. The kernel size $k_b$ is set to increase with the network depth $k$ ($k \in \{1, 2, 3, 4\}$ in our experiments). It is denoted as

$$\hat{X}' = \mathrm{DWConv}_{k_b \times 1}\left(\mathrm{DWConv}_{1 \times k_b}\left(\mathrm{Avg}\left(\hat{X}\right)\right)\right) \tag{16}$$

$$k_b = 2k + 3 \tag{17}$$

where $\hat{X}$ represents the input feature map, DWConv represents the depthwise separable convolution operation, and Avg represents the average pooling operation. Such a design allows the model to gradually increase the receptive field as the number of layers deepens. When the receptive field is small, it focuses on local details, such as textures and edges in RSIs. When the receptive field increases, it captures broader spatial structures and long-range dependencies, such as roads and rivers in RSIs.

*2) Frequency Domain:* Inspired by the wide application of frequency-domain analysis in image processing [56], [57], we introduce the fast Fourier transform (FFT) for global frequency-domain information modeling. Compared with traditional convolution operations that only model spatial-domain features, FFT allows us to analyze the global information of feature maps in the frequency domain, thereby separating high and low frequency components and providing richer feature representations for image details enhancement and structure preservation in RSIs. First, we independently apply a 2-D FFT (FFT2D) to each channel along the spatial dimensions of the input feature map

$$F_{\mathrm{FFT}} = \mathrm{FFT2}D\left(\hat{X}\right) \tag{18}$$

where the output $F_{\mathrm{FFT}}$ is a complex tensor. This transformation can break down the original spatial-domain features into different frequency-domain components, allowing the model to distinguish high-frequency information from low-frequency information more clearly. Subsequently, the differential frequency-domain feature is obtained by element-wise subtraction, which can be denoted as

$$F_C = F_{\mathrm{FFT1}} - F_{\mathrm{FFT2}}. \tag{19}$$

This difference features $F_C$, which preliminarily indicates which frequency-domain components have changed between the bi-temporal images. However, using only the difference may ignore the directionality or gradient of changed regions. Therefore, frequency-domain gradient information is introduced. If the gradient of a frequency component is large, it indicates that the frequency has undergone significant and localized changes between the bi-temporal images, and is more likely to be true changed regions (e.g., new construction). If the gradient is small, it is more likely to be pseudo-changed regions (e.g., gradual changes in lighting). Therefore, the gradient information of the frequency-domain differential features can enhance the model's sensitivity to abrupt changes and suppress gradual pseudo-changes.

The original frequency-domain difference features are concatenated with the calculated gradient information ($F_{\mathrm{grad}}$) to form an enhanced feature representation, which can be denoted as

$$M_X = \mathrm{Concat}\left(F_C, F_{\mathrm{grad}}\right). \tag{20}$$

Next, we fine-tune the amplitude spectrum to enhance the global interaction across channels

$$M'_X = \sigma\left(\mathrm{Conv}^{(2)}_{1\times1}\left(\mathrm{SiLU}\left(\mathrm{Conv}^{(1)}_{1\times1}\left(M_X\right)\right)\right)\right). \tag{21}$$

Based on the optimized difference features, a learnable attention mask $M'_X$ is generated to adaptively assign weights to different frequency components. The generated attention mask is multiplied element-wise with the original frequency-domain features to enhance frequency components related to changes and suppress those related to pseudo-changes, which is expressed as

$$F'_{\mathrm{FFT}} = M'_X \odot F_{\mathrm{FFT}}. \tag{22}$$

The result is then converted back to the spatial domain via the inverse FFT (IFFT)

$$\hat{X}_{\mathrm{FFT}} = \mathrm{IFFT2D}\left(F'_{\mathrm{FFT}}\right). \tag{23}$$

We only retain the real part of the IFFT output and discard the imaginary component. The result $\hat{X}_{\mathrm{FFT}}$ is added to the axially aware features $\hat{X}'$ for feature fusion, and the channel modulation feature map $A_c$ is calculated. Finally, we adjust the input features in a channel-by-channel weighting manner to obtain the output result $X'$

$$X_f = \hat{X}' + \hat{X}_{\mathrm{FFT}} \tag{24}$$

$$A_c = \mathrm{SiLU}\left(\mathrm{Conv}_{1\times1}\left(X_f\right)\right) \tag{25}$$

$$X' = A_c \odot \hat{X}. \tag{26}$$

Through a dual-path design of "axial modeling + spectral modulation," the SDM Module achieves local structural enhancement in the spatial domain and global spectral modulation in the frequency domain, respectively. Specifically, the axial branch employs strip convolutions to expand the receptive field, thereby strengthening the spatial continuity of strip-shaped changes. As a complementary path, the frequency branch performs global spectral modeling on the feature maps. It enhances edge-dominated high-frequency features and suppresses low-frequency background variations, thereby complementing the axial branch. It therefore makes up for the limitations of FastSAM in axial perception and frequency-domain separation.

## IV. Experimental Results

### A. Datasets

To verify the effectiveness of the proposed FastSAM-CD method, we conducted a comprehensive experimental evaluation on three publicly available RSICD benchmark datasets, including learning, vision and remote (LEVIR)-CD [23], Wuhan University (WHU)-CD [58], and China Land Cover Dataset (CLCD) [59], which, respectively, cover various types of changes in urban and suburban environments. The three datasets provide a rich array of test scenarios for the model, and the following is a detailed introduction to each of them.

*1) LEVIR-CD Dataset:* The bi-temporal images in this dataset are from 20 different regions in Texas, USA. The number of image pairs in the training set, validation set, and test set is 445, 64, and 128, respectively. Each image is $1024 \times 1024$ pixels in size and covers various types of buildings such as residences and warehouses.

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*2) WHU-CD Dataset:* This dataset contains more than 220 000 individual buildings in New Zealand extracted from aerial images within a 450 $km^2$ area, covering a 20.5 $km^2$ urban area, with images having a spatial resolution of 0.2 m per pixel. The number of image pairs in the training set, validation set, and testing set is 5948, 743, and 743, respectively, and each image is $256 \times 256$ pixels in size.

*3) CLCD Dataset:* This dataset is based on 335 709 Landsat images on Google Earth Engine. The number of image pairs in the training set, validation set, and testing set is 320, 120, and 120, respectively. The dataset mainly focuses on changes in cultivated land. It includes nine types of land use, such as farmland, forest, shrub, grassland, water area, and wetland.

### B. Implementation Details and Evaluation Metrics

We used precision (Pre), recall (Rec), $F1$-score ($F1$), intersection over union (IoU), and overall accuracy (OA) as the main evaluation metrics to quantify the performance of the model [60], and their formulas are as follows:

$$\begin{aligned} \text{Pre} &= \frac{\text{TP}}{\text{TP}+\text{FP}}, \quad \text{Rec} = \frac{\text{TP}}{\text{TP}+\text{FN}} \\ F_1 &= 2 \cdot \frac{\text{Pre} \cdot \text{Rec}}{\text{Pre}+\text{Rec}}, \quad \text{IoU} = \frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}} \\ \text{OA} &= \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}} \end{aligned} \tag{27}$$

where TP, FP, TN, and FN, respectively, represent the quantities of true positives, false positives, true negatives, and false negatives.

The proposed FastSAM-CD was based on the PyTorch framework and was trained for 50 epochs using an NVIDIA GeForce RTX 3090 GPU. In the experiment, images from the LEVIR-CD dataset and the CLCD dataset were cropped to $512 \times 512$, and images from the WHU-CD dataset were cropped to $256 \times 256$ as input for training the model. During the training process, we used random vertical flipping, random horizontal flipping, and random rotation with a random scale of [0.75, 1.0, 1.25], updated the parameters using the Adam optimizer, and set the initial learning rate to 0.001. To further optimize the training process, we implemented a learning rate adjustment strategy based on polynomial decay. The learning rate was adjusted dynamically according to the formula

$$\eta_t = \eta_0 \times \left(1 - \frac{t}{T}\right)^{3.0} \tag{28}$$

where $t$ represents the current number of completed iterations and $T$ represents the total number of iterations. On this basis, the actual running learning rate $\eta_t$ is determined by multiplying the initial learning rate $\eta_0$ by that ratio. This polynomial decreasing approach allows the model to converge rapidly at a higher learning rate in the early stages of training, and gradually decrease the learning rate toward the end of training to fine-tune the model parameters, thereby enhancing the model's generalization ability.

### C. Comparison With Supervised CD Methods

To demonstrate the superiority of the proposed FastSAM-CD method, we compare it with state-of-the-art (SOTA) RSICD methods, which can be roughly divided into five categories.

1) *Methods Based on the CNN:* FC-SD [29], FC-EF [29], STANet [23], DT-SCN [28], DSAMNet [26], EGRCNN [25], DMINet [24], and CCLNet [61].
2) *Methods Based on the Transformer:* BIT [30], ChangeFormer [31], MFATNet [32], TCIANet [33], and RFormer [34].
3) *Methods Based on a Hybrid Architecture of the CNN and Transformer:* CTFINet [35], HSSENet [36], and EATDer [37].
4) *Diffusion Model-Based Method:* DDPM-CD [62].
5) *Methods Based on SAM:* TTP [63], SAM-CD [45], and ASS-CD [46].

*Quantitative Evaluation:* We compared the accuracy of the proposed FastSAM-CD method with that of SOTA CD methods and reported the results of the different methods on the three benchmark datasets in Table I.

*1) Results on the LEVIR-CD Dataset:* The results show that the FastSAM-CD method scored 92.81% on the $F1$ metric, which is 2.03% higher than the CNN-based SOTA CD method EGRCNN [25]. The dataset has a wide range of small-scale targets and fine-grained boundary structures, such as change categories like "small buildings" and "road surfaces," which place higher demands on the model's capacity to capture local details, and FastSAM-CD performs better in these categories.

*2) Results on the WHU-CD Dataset:* Among all methods, our proposed FastSAM-CD achieves the best performance, with the $F1$ metric scoring 93.80%, and its $F1$ metric scores are improved by 3.23% and 3.05% compared to TTP [63] and SAM-CD [45], respectively. The dataset mainly consists of buildings with a wide range of target structures and strong directionality, which demands a high level of structural consistency modeling and axial perception ability for the model. FastSAM-CD performs well in this scenario, indicating that the model is better at discriminating strip-shaped boundaries and structural contours.

*3) Results on the CLCD Dataset:* The results show FastSAM-CD has the best $F1$-score of 89.79% among all methods. This further confirms the stability and reliability of FastSAM-CD across different datasets. Notably, the $F1$ metric score on the CLCD dataset decreased slightly compared to the results on other datasets, possibly because the CLCD dataset contains nine complex land types, specific types that the FastSAM encoder handles less frequently. Despite this, FastSAM-CD still maintains strong detection capabilities across different semantic distributions.

On three validated datasets, the proposed FastSAM-CD method achieves higher $F1$-scores. In the experimental results obtained on the LEVIR-CD dataset, the WHU-CD dataset, and the CLCD dataset, the $F1$-score is 1.29%, 1.72%, and 1.13% higher than that of SOTA CD methods, respectively, indicating that, driven by the HFM Adapter and SDM Module, FastSAM-CD has a stronger ability to extract image detail features and is better suited to the RSICD task.

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TABLE I

QUANTITATIVE COMPARISONS IN TERMS OF PRE (%), REC (%), $F1$ (%), AND IOU (%) ON THE THREE RSICD DATASETS. $F1$ AND IOU ARE COMPREHENSIVE METRICS; HIGHER $F1$ AND IOU VALUES INDICATE BETTER MODEL PERFORMANCE. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

| Method | Conference/Journal | LEVIR-CD | | | | WHU-CD | | | | CLCD | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Pre | Rec | F1 | IoU | Pre | Rec | F1 | IoU | Pre | Rec | F1 | IoU |
| FC-SD [29] | ICIP 2018 | 88.36 | 83.97 | 86.11 | 75.61 | 50.32 | 79.56 | 61.65 | 44.56 | 61.96 | 69.01 | 65.30 | 48.48 |
| FC-EF [29] | ICIP 2018 | 85.76 | 80.22 | 82.90 | 70.79 | 70.37 | 64.52 | 67.32 | 50.74 | 66.29 | 68.19 | 67.22 | 50.63 |
| STANet [23] | RS 2020 | 83.15 | 90.83 | 86.82 | 76.71 | 78.92 | 85.22 | 81.95 | 69.42 | 73.73 | 77.63 | 75.63 | 60.81 |
| DT-SCN [28] | TGRS 2020 | 88.17 | 85.92 | 87.03 | 77.04 | 63.76 | 83.11 | 72.16 | 56.45 | 76.37 | 80.92 | 78.59 | 64.73 |
| EGRCNN [25] | TGRS 2022 | 90.02 | 91.55 | 90.78 | 83.12 | 90.22 | 88.59 | 89.40 | 80.83 | 83.86 | 81.88 | 82.87 | 70.75 |
| DSAMNet [26] | TGRS 2022 | 81.95 | 88.81 | 85.24 | 74.28 | 73.84 | 88.94 | 80.69 | 67.63 | 81.15 | 81.61 | 81.38 | 68.61 |
| DMINet [24] | TGRS 2023 | 91.23 | 88.76 | 89.98 | 81.79 | 93.92 | 83.01 | 88.13 | 78.78 | 86.23 | 88.77 | 87.48 | 77.75 |
| CCLNet [61] | TGRS 2023 | 90.76 | 89.47 | 90.11 | 82.00 | 86.64 | 89.36 | 87.98 | 78.54 | 85.64 | 86.75 | 86.19 | 75.73 |
| BIT [30] | TGRS 2021 | 89.01 | 88.85 | 88.93 | 80.07 | 85.22 | 80.08 | 82.57 | 70.31 | 87.58 | 86.84 | 87.21 | 77.32 |
| ChangeFormer [31] | IGARSS 2022 | 90.65 | 88.97 | 89.80 | 81.49 | 88.18 | 88.44 | 88.31 | 79.07 | 81.80 | 82.26 | 82.03 | 69.53 |
| MFATNet [32] | RS 2022 | 91.12 | 88.99 | 90.04 | 81.88 | 91.45 | 84.63 | 87.91 | 78.43 | 82.97 | 81.95 | 82.46 | 70.15 |
| TCIANet [33] | JSTARS 2023 | 91.88 | 90.65 | 91.26 | 83.92 | 88.89 | 81.51 | 85.04 | 73.97 | 89.69 | 81.41 | 85.35 | 74.44 |
| RFormer [34] | TGRS 2024 | 91.68 | 89.47 | 90.56 | 82.75 | 92.42 | 91.25 | 91.83 | 84.89 | 86.36 | 85.96 | 86.16 | 75.69 |
| CTFINet [35] | ICASSP 2023 | 91.27 | 89.61 | 90.43 | 82.53 | 89.52 | 81.55 | 85.35 | 74.44 | 80.96 | 87.75 | 84.22 | 72.74 |
| HSSENet [36] | TGRS 2023 | 92.08 | 90.18 | 91.12 | 83.69 | 93.23 | 88.91 | 91.02 | 83.52 | 87.54 | 84.86 | 86.18 | 75.72 |
| EATDer [37] | TGRS 2024 | 91.88 | 90.45 | 91.16 | 83.76 | 90.74 | 89.00 | 89.86 | 81.59 | 86.73 | 87.11 | 86.92 | 76.87 |
| DDPM-CD [62] | Arxiv 2022 | 90.98 | 89.63 | 90.30 | 82.32 | 92.48 | 91.49 | 91.98 | 85.15 | 88.10 | 84.59 | 86.31 | 75.92 |
| TTP [63] | IGARSS 2024 | 92.35 | 90.71 | 91.52 | 84.37 | 94.69 | 86.79 | 90.57 | 82.76 | 85.64 | 86.74 | 86.19 | 75.73 |
| SAM-CD [45] | TGRS 2024 | 90.82 | 89.46 | 90.14 | 82.05 | 93.91 | 87.80 | 90.75 | 83.07 | 87.41 | 89.95 | 88.66 | 79.63 |
| ASS-CD [46] | RS 2025 | 91.56 | 89.92 | 90.73 | 83.03 | 91.24 | 92.94 | 92.08 | 85.32 | 87.54 | 86.55 | 87.04 | 77.05 |
| FastSAM-CD (Ours) | | 91.78 | 93.86 | 92.81 | 86.58 | 92.51 | 95.13 | 93.80 | 88.32 | 89.56 | 90.02 | 89.79 | 81.47 |

*a) Qualitative Evaluation:* As shown in Fig. 6, we present a visual comparison of FastSAM-CD with other mainstream methods on the three RSICD datasets.

*4) Results on the LEVIR-CD Dataset:* As shown in Fig. 6 (LEVIR-CD), the change area is mainly composed of small-scale densely arranged buildings with complex background textures, which is a typical scene of high density change and strong interclass similarity. FastSAM-CD can effectively reduce FP areas due to background noise and maintain a relatively complete target boundary profile. In contrast, methods such as BIT have more false detections or edge breaks.

*5) Results on the WHU-CD Dataset:* As shown in Fig. 6 (WHU-CD), the change targets are large-scale buildings with relatively less background interference, which is suitable for evaluating the model's ability to perceive cross-scale spatial structures. FastSAM-CD outperforms existing methods in terms of predictive accuracy for target boundaries, with the fewest FP and FN regions, demonstrating stronger structure-preserving ability.

*6) Results on the CLCD Dataset:* As shown in Fig. 6 (CLCD), the changed targets include complex structures, such as roads, water bodies, and farmland, showing significant directional and texture-heterogeneous properties. FastSAM-CD is the only model that can continuously predict the structure of slender roads, while other methods show structural fractures, demonstrating FastSAM-CD's superiority in capturing strip-shaped target changes.

FastSAM-CD demonstrates superior visualization results in multiple dimensions, including small-target detection, strip-shaped structure modeling, and complex background suppression.

### D. Comparison With Semi-Supervised CD Methods

In the experiment, we find that although FastSAM-CD is a fully supervised CD method, it achieves strong performance even with limited labeled training data. We compared FastSAM-CD with several semi-supervised CD methods, including AdvNet [64], SemiCDNet [65], and s4GAN [66]. In the semi-supervised setup, we randomly selected the required amount of labeled data from the original training set and treated the remaining training images as unlabeled data by discarding the corresponding change labels. All semi-supervised methods are trained under the same labeled-unlabeled split and unlabeled data protocol. FastSAM-CD is trained only on the labeled subset corresponding to each label ratio, and the remaining images are not used for the training process.

*1) Quantitative Evaluation:* We used IoU and OA as the primary performance metrics to quantify the model's performance. The quantitative results (see Table II) indicate that FastSAM-CD outperforms most semi-supervised RSICD

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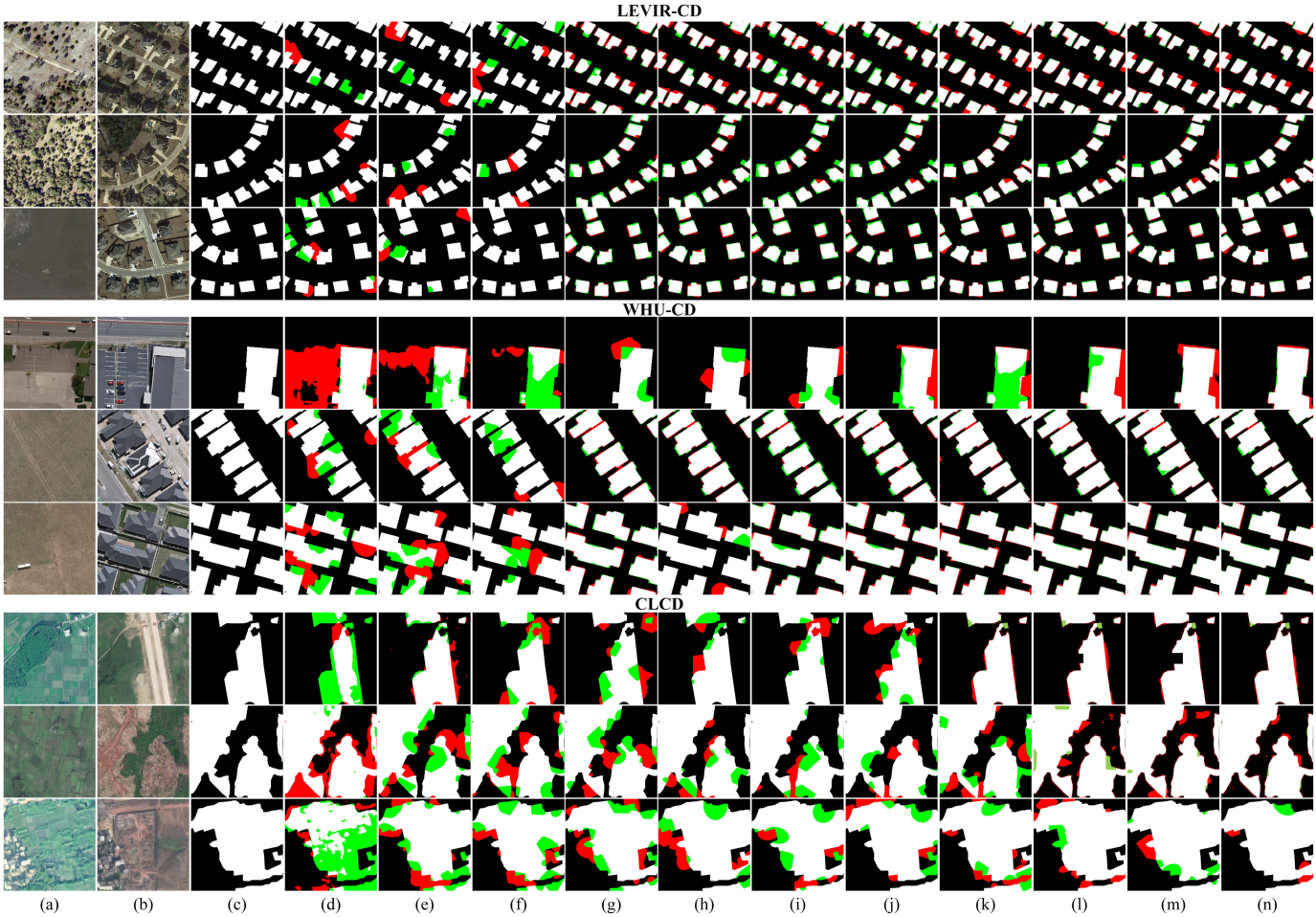



Fig. 6. Visualization comparison of results on three datasets. (a) T1 images. (b) T2 images. (c) Ground truth. (d) EGRCNN. (e) CCLNet. (f) BIT. (g) ChangeFormer. (h) RFormer. (i) HSSENet. (j) EATDer. (k) DDPM-CD. (l) TTP. (m) SAM-CD. (n) FastSAM-CD (ours). The rendered colors represent true positives (white), false positives (red), true negatives (black), and false negatives (green).

TABLE II

ACCURACY OF THE FASTSAM-CD METHOD AND SEMI-SUPERVISED CD METHODS OBTAINED ON DATASETS WITH DIFFERENT PROPORTIONS OF TRAINING DATA (%)

| Method | LEVIR-CD (% labeled) | | | | | | | | WHU-CD (% labeled) | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 5% | | 10% | | 20% | | 40% | | 5% | | 10% | | 20% | | 40% | |
| | IoU | OA | IoU | OA | IoU | OA | IoU | OA | IoU | OA | IoU | OA | IoU | OA | IoU | OA |
| AdvNet [64] | 66.13 | 98.08 | 72.39 | 98.45 | 74.60 | 98.58 | 75.01 | 98.60 | 56.12 | 97.92 | 61.60 | 98.64 | 73.82 | 98.80 | 75.82 | 98.80 |
| SemiCDNet [65] | 67.66 | 98.17 | 71.58 | 98.42 | 74.34 | 98.58 | 75.55 | 98.63 | 54.17 | 97.17 | 62.66 | 98.51 | 67.43 | 98.46 | 68.46 | 98.46 |
| s4GAN [66] | 64.02 | 97.89 | 67.01 | 98.11 | 73.42 | 98.51 | 75.44 | 98.62 | 58.22 | 96.89 | 63.42 | 98.42 | 70.58 | 98.80 | 76.80 | 98.80 |
| FastSAM-CD (Ours) | 66.54 | 97.92 | 80.60 | 98.67 | 82.24 | 98.70 | 82.98 | 98.73 | 57.12 | 97.32 | 70.70 | 98.87 | 79.99 | 99.10 | 83.42 | 99.26 |

methods, achieving the highest IoU score when using 10%, 20%, and 40% of the training data. However, it does not achieve the highest IoU score when using 5% of the training data. This is because training the model still requires a certain number of labeled samples to fit the RSICD task.

### E. Ablation Analysis

*1) Effectiveness of the Proposed Components:* In this ablation study, the baseline used two Siamese ResNets to extract features. Starting from the baseline, we gradually incorporated the FastSAM encoder, HFM Adapter, and SDM Module into the model network. All configurations reported in Table III are trained with the same data augmentation strategy and training schedule described in Section IV-B.

*a) Quantitative evaluation:* As shown in Table III, the use of the FastSAM encoder significantly improves the detection accuracy compared with the baseline method, with $F1$ and IoU scores increasing by 3.70% and 5.89%, respectively. When the HFM Adapter and the SDM Module are added to the encoder based on the FastSAM, $F1$ and IoU scores increase again by 4.05% and 6.91%, respectively.

*b) Qualitative evaluation:* As shown in Fig. 7, there are obvious FP and FN regions in the prediction results of the

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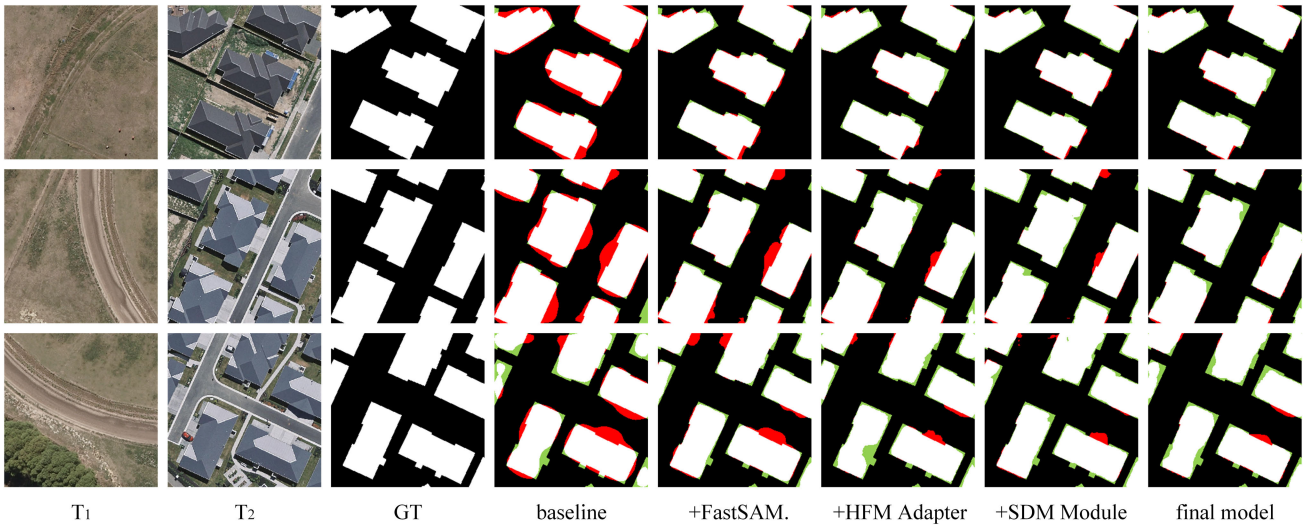



Fig. 7. Visualization comparison of results in the ablation study. The rendered colors represent true positives (white), false positives (red), true negatives (black), and false negatives (green).

TABLE III
PERFORMANCE COMPARISON UNDER DIFFERENT CONFIGURATIONS (TESTED ON WHU-CD) (%)

| Configuration | F1 | IoU | OA |
|---|---|---|---|
| Baseline | 86.05 | 75.52 | 98.75 |
| FastSAM | 89.75 | 81.41 | 98.98 |
| FastSAM + HFM Adapter | 92.65 | 86.31 | 99.12 |
| FastSAM + SDM Module | 91.82 | 84.88 | 99.01 |
| FastSAM + HFM Adapter + SDM Module | 93.80 | 88.32 | 99.08 |

TABLE IV
ROBUSTNESS OF DIFFERENT VARIANTS TO INPUT PERTURBATIONS (TESTED ON LEVIR-CD) (%)

| Type | Setting | Baseline | | SDM-spatial | | FastSAM-CD | |
|---|---|---|---|---|---|---|---|
| | | F1 | IoU | F1 | IoU | F1 | IoU |
| - | - | 85.41 | 74.53 | 89.65 | 81.90 | 92.81 | 86.58 |
| Registration | $\Delta x = 5$ | 82.83 | 70.69 | 87.46 | 77.71 | 90.95 | 83.40 |
| | $\Delta y = 5$ | 82.32 | 69.96 | 87.11 | 77.10 | 90.65 | 82.90 |
| Noise | $\sigma = 0.01$ | 83.55 | 71.74 | 88.44 | 79.27 | 91.94 | 85.09 |
| | $\sigma = 0.03$ | 83.52 | 71.71 | 88.23 | 78.93 | 91.77 | 84.79 |
| Illumination | $\alpha = 0.8$ | 83.09 | 71.08 | 87.81 | 78.10 | 91.44 | 84.24 |
| | $\alpha = 1.2$ | 82.94 | 70.98 | 87.58 | 77.91 | 91.24 | 83.90 |

baseline, and the target contours are missing or misjudged, indicating low overall recognition accuracy. With the introduction of FastSAM, the contour structure of the target area can be captured more accurately, and the number of false detection areas is significantly reduced, indicating that the basic segmentation ability of the model has been improved. With the further integration of the HFM Adapter, the model's ability to depict the edges and details of the target has improved significantly, especially in the complex structures (see Fig. 7, rows 2 and 3), where it can more clearly distinguish the changed boundaries from the background interference. The visualizations after integrating the SDM Module show that the model has been further optimized in terms of structural continuity and boundary fit. The final FastSAM-CD model shows the closest prediction performance to GT in all three samples, with the least error and the clearest boundaries.

*2) Effectiveness of the SDM Module Under Input Perturbations:* To further verify the effectiveness of the proposed SDM Module, we evaluate the robustness of different model variants under input perturbations. We introduce pixel-level translations between the bi-temporal images to simulate registration errors, where $\Delta x$ and $\Delta y$ denote the horizontal and vertical pixel shifts, respectively. We also add Gaussian noise into bi-temporal images, with $\sigma$ representing the noise standard deviation. In addition, we apply a global intensity scaling to bi-temporal images to simulate illumination variations, where $\alpha$ is the scaling factor. Under these settings, we evaluate the baseline, the variant using only the spatial-domain branch of the SDM Module, and FastSAM-CD.

As summarized in Table IV, the FastSAM-CD consistently achieves the highest $F1$ and IoU scores across all perturbation settings. More importantly, as the perturbation strength increases, FastSAM-CD degrades more slowly than the baseline and the spatial-only variant, indicating that the frequency-domain branch in the SDM Module enhances robustness to realistic imaging distortions in CD.

### F. Discussion of Architecture and Model Efficiency

*1) Computational Cost:* Table V presents the computational resource costs of various CD methods. It is observed that while the number of parameters (Params) and floating-point operations (FLOPs) significantly impact model performance, they are not the sole determining factors. Among all the methods compared, BIT [30] has the highest inference speed. BAN [67] has the fewest parameters. In FastSAM-CD, the FastSAM encoder takes up the majority of parameters (68M), while the proposed network module takes up a total of 6.5M parameters. FastSAM-CD demonstrates competitive computational efficiency and moderate inference speed. We believe that FastSAM-CD strikes a good balance between performance and efficiency, making it particularly suitable for scenarios with limited data and computational resources.

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TABLE V
COMPUTATIONAL COSTS OF DIFFERENT CD METHODS

| Methods | Params (M) | FLOPs (G) | Infer. (s) |
|---|---|---|---|
| BIT [30] | 6.93 | 8.44 | 1.12 |
| DDPM-CD [62] | 46.41 | 742.48 | 30.6 |
| DSAMNet [26] | 16.95 | 75.29 | 5.26 |
| EATDer [37] | 6.61 | 23.46 | 3.67 |
| RFormer [34] | 21.11 | 11.36 | 5.52 |
| BAN [67] | 3.80 | 330.32 | - |
| SFCD-Net [68] | 14.56 | 273.43 | - |
| Meta-CD [69] | 13.67 | 10.42 | 5.64 |
| FastSAM-CD (Ours) | 74.50 | 8.68 | 4.46 |

TABLE VI
MODEL PERFORMANCE WITH DIFFERENT NUMBER OF HEADS IN MULTISCALE SPATIAL CONVOLUTION OF THE HFM ADAPTER (TESTED ON WHU-CD) (%)

| Number of Heads | Pre | Rec | F1 | IoU | OA |
|---|---|---|---|---|---|
| 1 | 92.05 | 91.65 | 91.85 | 84.93 | 98.47 |
| 2 | 92.14 | 91.88 | 92.01 | 85.20 | 98.87 |
| 4 | 92.51 | 95.13 | 93.80 | 88.32 | 99.08 |
| 8 | 92.99 | 92.77 | 92.88 | 86.71 | 99.12 |
| 16 | 89.17 | 88.75 | 88.96 | 80.12 | 98.55 |

*2) Number of Convolutional Heads in the HFM Adapter:* When splitting the input feature into multiple parts in the channel dimension, we consider whether the number of convolutional heads affects the feature extraction ability and thereby the model performance. We conducted experiments to select the optimal number of convolutional heads, and the results are reported in Table VI.

As can be observed from Table VI, relatively high $F1$ and IoU scores can be achieved when performing CD tasks with four convolutional heads, which is in line with our intuition that an appropriate number of convolutional heads can better capture multiscale features while maintaining high computational efficiency. Also, as the results of using one and eight convolutional heads show, too few or too many convolutional heads can lead to a decline in experimental results. When only one convolutional head is used, the model has difficulty fully capturing complex multiscale features. When using eight convolutional heads, although the model can capture more details, it introduces redundant information due to the increased computational complexity, resulting in a decline in model performance.

## V. CONCLUSION

In this article, we propose a FastSAM-CD framework for RSICD tasks. Based on the strong feature extraction capabilities of FastSAM, this framework designs two collaborative optimization modules to enhance task adaptability for the multiscale heterogeneity characteristics in RSIs. First, the designed HFM Adapter jointly models from both the local–global and spatial–channel dimensions, enhancing the encoder's ability to model cross-scale varying features. It also addresses the problem of FastSAM's insufficient feature extraction ability in the encoding stage due to semantic misalignment and geometric misalignment in RSIs. Second, the SDM Module combines spectral decomposition and axial sensing mechanisms to dynamically adjust the channel response, enhancing the decoder's ability to represent direction-sensitive and spectrally heterogeneous ground objects. The results demonstrate that FastSAM-CD achieves excellent performance on multiple RSICD datasets.

However, there are several limitations of FastSAM-CD. First, because the framework builds on the FastSAM backbone that is designed for natural images, it may still suffer from a domain gap when applied to RSIs. For example, the performance on the CLCD dataset decreases slightly for complex cultivated land types that appear less frequently in the pretraining data. Second, all experiments are conducted on three optical RSICD datasets, which mainly involve building and cultivated land changes. Therefore, the generalization of FastSAM-CD to other sensors (e.g., SAR, multispectral, and hyperspectral) and to broader geographic areas remains unverified.

In the future, an important research trend in RSICD is to integrate VFMs with multimodal, semantic-level CD. By jointly leveraging heterogeneous data, high-level semantics, and text prompt information, these models are expected to not only localize where changes occur but also describe what has changed in practical scenarios. Meanwhile, diffusion models and other generative methods can be combined with VFMs to synthesize change samples, thereby supporting model training and evaluation in complex scenes. In addition, VFM-based CD methods can be further incorporated into pipelines that balance privacy preservation and lightweight deployment. For example, it is significant to study cross-institution federated learning and explore more efficient training and deployment strategies, which can meet practical requirements on computational efficiency, data security, and reliability in remote sensing applications.

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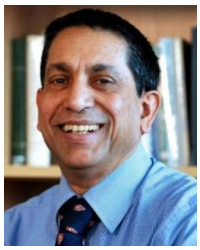
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